

Virus Dynamics in R

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Preface

This is an attempt to reproduce the code from *Virus Dynamics: Mathematical Principles of Immunology and Virology* by Nowak and May (Nowak and May 2000).

No guarantees that anything in this is correct; it is just an attempt to reproduce what is in the book.

This book is written in Quarto markdown (.qmd) and rendered using the phenomenal Quarto engine.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Introduction

This is a book created from markdown and executable code.

2 The Basic Model of Virus Dynamics

```
library(deSolve)
library(tidyverse)
```

The basic model of virus dynamics (Nowak and May 2000) tracks three populations: uninfected target cells x , infected cells y , and free virus particles v . Uninfected cells are produced at a constant rate λ , die at rate d , and become infected on contact with virus at rate β . Infected cells die at rate a , and each produces free virus at rate k , which is in turn cleared at rate u .

$$\begin{aligned}\frac{dx}{dt} &= \lambda - dx - \beta xv \\ \frac{dy}{dt} &= \beta xv - ay \\ \frac{dv}{dt} &= ky - uv\end{aligned}$$

The basic reproductive ratio of the virus, the average number of newly infected cells arising from one infected cell placed in a fully susceptible cell population, is

$$R_0 = \frac{\beta \lambda k}{adu}$$

The virus can establish an infection only if $R_0 > 1$.

2.1 Set Up the ODE

```
base_ode <- function(time, state, parameters){
  with(as.list(c(state, parameters)),{

    dx <- lambda - d*x - beta * x * v
    dy <- beta * x * v - a * y
    dv <- k * y - u * v
```

```

    return(list(c(dx,dy,dv)))
  })
}

t <- seq(0,30,.1)

params <- c(
  lambda = 1e5, # Uninfected cell production rate
  d = .1, # Cell Death Rate
  a = .5, # Infected Cell Death Rate
  beta = 2e-7, # "Rate Constant"
  k = 100, # Virus production from infected cell
  u = 5 # Free virus lifespan
)

```

2.2 Solve the System

Guessing initial values from a graph:

```

x0 <- params["lambda"][1]/params["d"][1]
init <- c(x = unname(x0),
          y = 1, v = 1)

out <- ode(init, t, base_ode, params)

out_df <- as_tibble(as.data.frame(out))

```

Note that the differences in the scales across compartments are reproduced almost immediately, as in the book.

```

compartment_names <- c("Uninfected Cells",
  "Infected Cells",
  "Free Virus")

p <- out_df %>%
  setNames(c("time", compartment_names)) %>%
  gather(compartment, value, -time) %>%
  mutate(compartment = factor(compartment, compartment_names)) %>%
  ggplot(aes(time, y = value))+

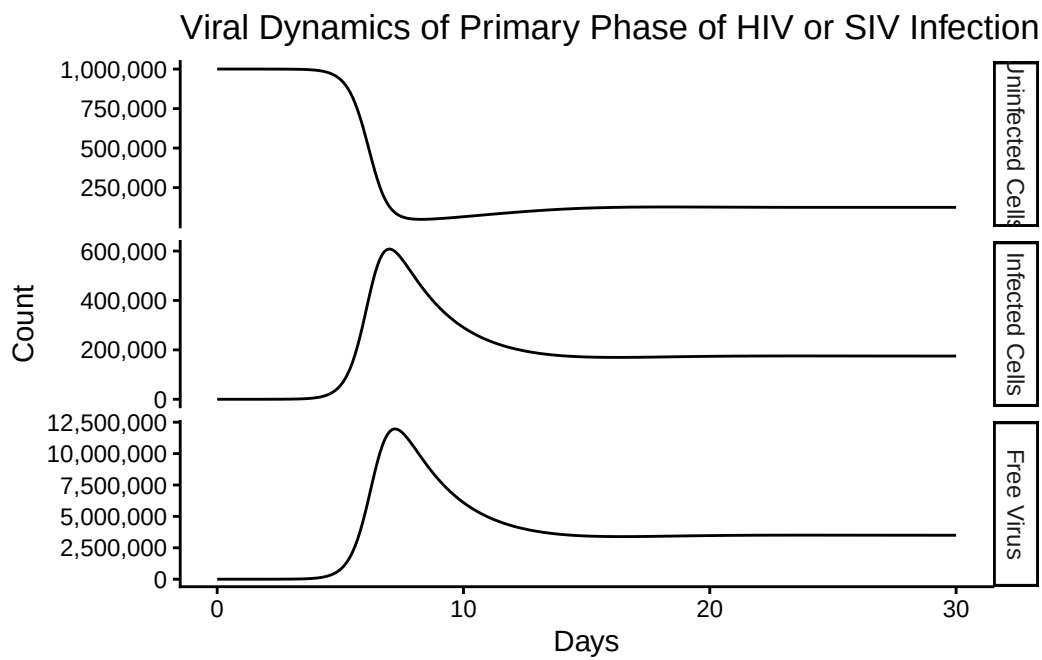
```

```

geom_line()+
facet_grid(rows = vars(compartment),scales = "free_y")+
theme_classic()+
labs(
  title = "Viral Dynamics of Primary Phase of HIV or SIV Infection",
  x = "Days",
  y = "Count"
)+
scale_y_continuous(labels = scales::comma_format(accuracy = 1000))

```

p



3 Anti-viral Drug Therapy

From pages 35-37.

```
library(deSolve)
library(tidyverse)
```

3.1 Set Up the ODE

Represents infected cells becoming:

- Latently infected: do not produce new virions, but contain replication competent virus that can be re-activated to become virus producing
- long-lived chronic producers: produce small amounts of virus over long periods
- cells which harbor defective provirus

These dynamics can be represented by the below equations:

$$\begin{aligned}\frac{dx}{dt} &= \lambda - dx - \beta xv \\ \frac{dy_1}{dt} &= q_1 \beta xv - a_1 y_1 + \alpha y_2 \\ \frac{dy_2}{dt} &= q_2 \beta xv - a_2 y_2 - \alpha y_2 \\ \frac{dy_3}{dt} &= q_3 \beta xv - a_3 y_3 \\ \frac{dv}{dt} &= ky_1 - uv\end{aligned}$$

These equations can then be coded as follows:


```

base_ode <- function(time, state, parameters){
  with(as.list(c(state, parameters)),{

    dx <- lambda - d*x - beta * x * v
    dy1 <- q1*beta * x * v - a1 * y1 + alpha * y2
    dy2 <- q2*beta * x * v - a2 * y2 - alpha * y2
    dy3 <- q3*beta * x * v - a3 * y3
    dv <- k * y1 - u * v

    return(list(c(dx, dy1, dy2, dy3, dv)))
  })
}

```

Now we can set up the initial conditions and constants to simulate the outputs from this system of equations.

```

t <- seq(0,30,.1)

params <- c(
  lambda = 1e7, # Uninfected cell production rate
  d = .1, # Cell Death Rate
  a1 = .5, # Infected Cell Death Rate
  a2 = .01, # Infected Cell Death Rate
  a3 = .008, # Infected Cell Death Rate
  beta = 5e-10, # "Rate Constant"
  alpha = .3, # Virus Production rate of reactivated latent
  k = 500, # Virus production from infected cell
  u = 5, # Free virus lifespan
  q1 = .55, # P(Infected State | Infected)
  q2 = .05, # P(Latent State | Infected)
  q3 = .4 # P(Defective Provirus | Infected)
)

#' Guessing Initial values from a graph
x0 <- params["lambda"][1]/params["d"][1]
init <- c(x = unname(x0),
         y1 = 1, y2=0, y3=0, v = 1)

out <- ode(init, t, base_ode, params)

out_df <- as_tibble(as.data.frame(out))

```

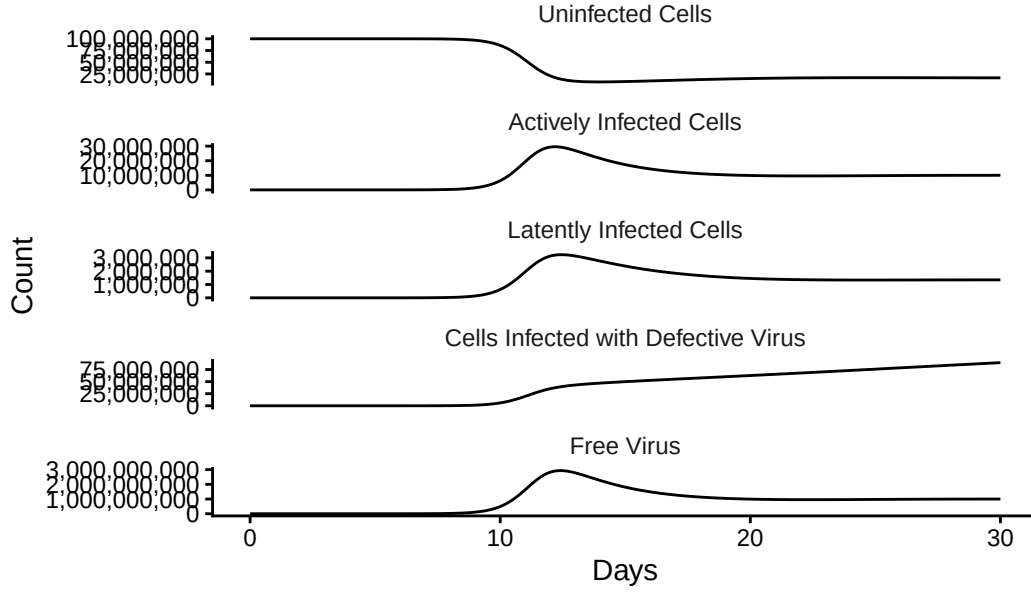
As before, the differences in the scales across compartments are reproduced almost immediately, as in the book.

```
compartment_names <- c("Uninfected Cells",
"Actively Infected Cells",
"Latently Infected Cells",
"Cells Infected with Defective Virus",
"Free Virus")

p <- out_df %>%
  setNames(c("time", compartment_names)) %>%
  gather(compartment, value, -time) %>%
  mutate(compartment = factor(compartment,
                              compartment_names)) %>%
  ggplot(aes(time, y = value))+
  geom_line()+
  facet_wrap(~compartment, ncol = 1, scales = "free_y")+
  theme_classic()+
  labs(
    title = "Viral Dynamics of Primary Phase of HIV or SIV Infection",
    x = "Days",
    y = "Count"
  )+
  scale_y_continuous(labels = scales::comma_format(accuracy = 1000))+
  theme(strip.background = element_blank())

p
```

Viral Dynamics of Primary Phase of HIV or SIV Infection



3.1.1 Equilibrium Conditions

$$\hat{x} = \frac{x_0}{R_0}$$

$$\hat{y}_1 = (R_0 - 1) \frac{du}{\beta k} = \hat{v} \frac{u}{k}$$

$$\hat{y}_2 = \frac{\hat{y}_1 \frac{a_1}{q_1}}{\frac{\alpha + a_2}{q_2} + \frac{\alpha}{q_1}}$$

$$\hat{y}_3 = \hat{y}_2 \frac{\frac{\alpha + a_2}{q_2}}{\frac{a_3}{q_3}}$$

$$\hat{v} = (R_0 - 1) \frac{d}{\beta}$$

With R_0 given by:

$$R_0 = \frac{\beta \lambda k}{a_1 du} \left(q_1 + q_2 \frac{\alpha}{\alpha + a_2} \right)$$

3.2 Effect of Anti-Virals

The big take-away from these equations is that introduction of an anti-viral should reduce the contact rate, β , to zero (or some much lower value).

```
params["beta"] <- 0

init <- c(x = unname(x0),
         y1 = 1000, y2=1000, y3=10000, v = 50)

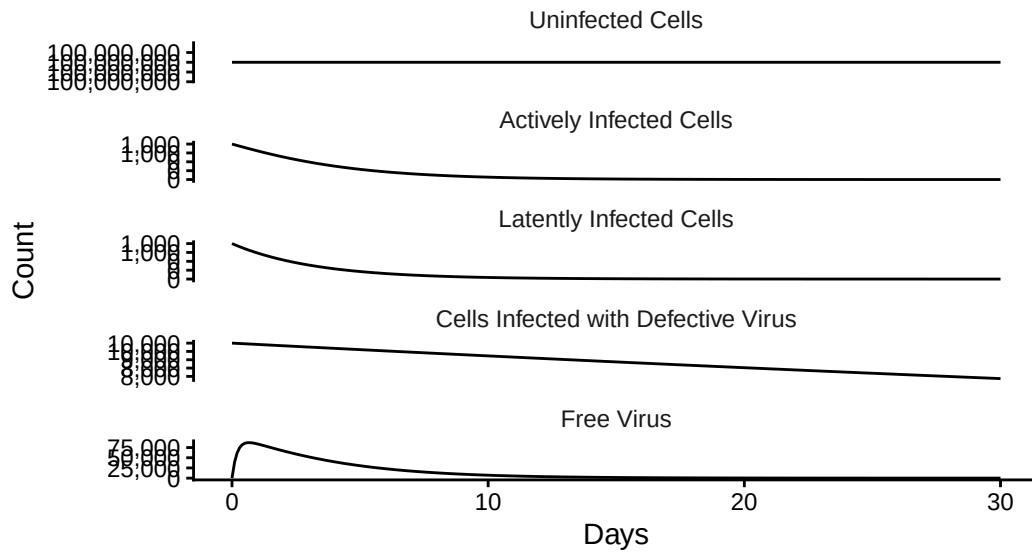
out_antiviral <- ode(init, t, base_ode, params)

out_df_av <- as_tibble(as.data.frame(out_antiviral))
```

```
p <- out_df_av %>%
  setNames(c("time", compartment_names)) %>%
  gather(compartment, value, -time) %>%
  mutate(compartment = factor(compartment,
                              compartment_names)) %>%
  ggplot(aes(time, y = value))+
  geom_line()+
  facet_wrap(~compartment, ncol = 1, scales = "free_y")+
  theme_classic()+
  labs(
    title = "Viral Dynamics of Primary Phase of HIV or SIV Infection",
    subtitle = "With Anti-viral therapy",
    x = "Days",
    y = "Count"
  )+
  scale_y_continuous(labels = scales::comma_format(accuracy = 1000))+
  theme(strip.background = element_blank())

p
```

Viral Dynamics of Primary Phase of HIV or SIV Infection With Anti-viral therapy



4 Dynamics of Hepatitis B Virus

More to come.

5 Dynamics of Immune Response

Nowak and May (Nowak and May 2000) argue that:

- (1) virus load is an important determinant of disease
- (2) immune responses limit virus load
- (3) individual variation in immune responsiveness accounts for much of the observed variation in the outcome of a disease

5.1 Predator-Prey Dynamics of the Immune System

We extend the basic model with a cytotoxic T lymphocyte (CTL) response z , which kills infected cells at rate p . In the simplest case the CTL are produced at a constant rate c and decay at rate b , independent of antigen.

$$\begin{aligned}\frac{dx}{dt} &= \lambda - dx - \beta xv \\ \frac{dy}{dt} &= \beta xv - ay - pyz \\ \frac{dv}{dt} &= ky - uv \\ \frac{dz}{dt} &= c - bz\end{aligned}$$

With the basic reproduction number defined for this system as

$$R_0 = \frac{\beta \lambda k}{adu}$$

And the basic reproduction number in the presence of an immune response (CTL)

$$R_1 = \frac{\beta \lambda k}{(a + a')du}$$

Where:

$$a' = \frac{cp}{b}$$

representing the rate at which infected cells are eliminated by the CTL response at equilibrium.

If $R_1 < 1$ then the infection will clear. The virus may spread initially, but once the immune response is activated, each infected cell will on average give rise to less than one newly infected cell and thus the infection will die out.

5.2 Simulating the Immune Response

```
library(deSolve)
library(tidyverse)
```

We code the four-compartment system and reuse the contact and replication parameters from the basic model, adding the CTL killing rate p , production rate c , and decay rate b .

```
immune_ode <- function(time, state, parameters){
  with(as.list(c(state, parameters)),{

    dx <- lambda - d * x - beta * x * v
    dy <- beta * x * v - a * y - p * y * z
    dv <- k * y - u * v
    dz <- c - b * z

    return(list(c(dx, dy, dv, dz)))
  })
}

t <- seq(0, 30, .1)

params <- c(
  lambda = 1e5, # Uninfected cell production rate
  d = .1,      # Cell death rate
  a = .5,      # Infected cell death rate
  beta = 2e-7, # Rate constant
  k = 100,     # Virus production from infected cell
  u = 5,       # Free virus lifespan
  p = 1,       # CTL killing rate
  c = .1,      # CTL production rate
```



```

  b = .05      # CTL decay rate
)

```

The constant-production assumption gives an equilibrium CTL level of $\hat{z} = c/b$, so we can compute the two reproductive ratios directly and confirm that the immune response lowers, but does not here eliminate, viral spread.

```

a_prime <- with(as.list(params), c * p / b)

R0 <- with(as.list(params), beta * lambda * k / (a * d * u))
R1 <- with(as.list(params), beta * lambda * k / ((a + a_prime) * d * u))

c(R0 = R0, R1 = R1)

```

```

R0  R1
8.0 1.6

```

```

x0 <- unname(params["lambda"] / params["d"])
init <- c(x = x0, y = 1, v = 1, z = unname(params["c"] / params["b"]))

out <- ode(init, t, immune_ode, params)
out_df <- as_tibble(as.data.frame(out))

```

```

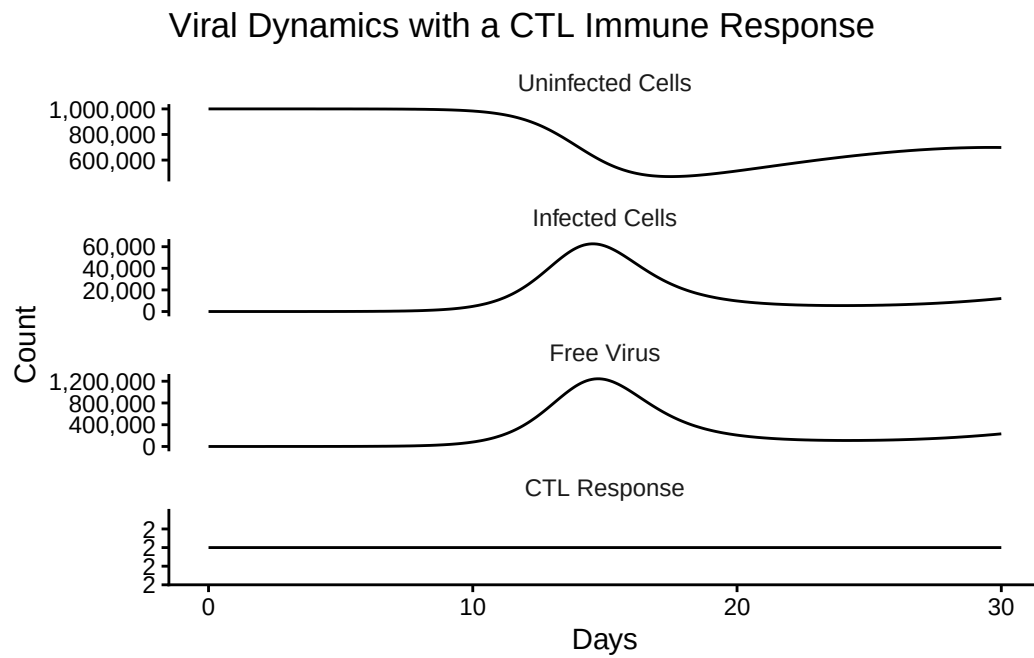
compartment_names <- c("Uninfected Cells",
                       "Infected Cells",
                       "Free Virus",
                       "CTL Response")

p_plot <- out_df %>%
  setNames(c("time", compartment_names)) %>%
  gather(compartment, value, -time) %>%
  mutate(compartment = factor(compartment, compartment_names)) %>%
  ggplot(aes(time, y = value)) +
  geom_line() +
  facet_wrap(~compartment, ncol = 1, scales = "free_y") +
  theme_classic() +
  labs(
    title = "Viral Dynamics with a CTL Immune Response",
    x = "Days",
    y = "Count"
  ) +

```

```
scale_y_continuous(labels = scales::comma_format(accuracy = 1)) +
theme(strip.background = element_blank())
```

p_plot



6 Summary

In summary, there will be more to come.

References

Nowak, Martin A., and Robert M. May. 2000. *Virus Dynamics: Mathematical Principles of Immunology and Virology*. Oxford University Press.